

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Mental Math (Gemini)

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[4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf](#)

Quadrant Check 1. In which two quadrants will you find the solutions for the basic trigonometric equation $\sin \theta = -\frac{\sqrt{3}}{2}$?

Finding All Solutions 2. When finding "all solutions" for an equation like $\tan \theta = -10$, what integer multiple do you add to your initial period solutions?

Zero-Product Property 3. If you apply the Zero-Product Property to the equation $\sec \theta (2 \cos \theta - \sqrt{2}) = 0$, what two basic equations are you left to solve?

Pythagorean Identity Substitution 4. What standard identity should you substitute for $\sec^2 \theta$ to convert the equation $2 \tan \theta + \sec^2 \theta = 4$ entirely into terms of tangent?

Isolating the Function 5. What is the isolated value of $\sin(3\theta)$ in the equation $2 \sin(3\theta) + 1 = 0$?

Addition/Subtraction Formulas 6. Using a trigonometric subtraction formula, what single function does the left side of the equation $\sin(2\theta) \cos(\theta) - \cos(2\theta) \sin(\theta) = \frac{\sqrt{3}}{2}$ simplify to?

Factoring Setup 7. To solve $3 \tan^3 \theta = \tan \theta$ by factoring, what is the required first step before you can factor out the greatest common factor?

Reciprocal Identities 8. If you multiply both sides of the equation $\sec(\frac{\theta}{2}) = \cos(\frac{\theta}{2})$ by $\cos(\frac{\theta}{2})$, what simplified equation do you get?

Solving Multiple Angles 9. If your intermediate step for solving a multiple-angle equation is $3\theta = \frac{3\pi}{2}$, what is the final value for θ ?

Matching Terms 10. To solve $\csc^2 \theta = \cot \theta + 3$ using algebraic factoring, what expression must replace $\csc^2 \theta$ so the equation is exclusively in terms of cotangent?